

Capstone Paper: Cross-Modal Representational Alignment in Vision and Audio Models

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Abstract

Representations in large language models (LLMs) have been shown to align with both vision and audio models (Huh et al., 2024; Wang et al., 2025), though such alignment may reflect shared semantic descriptions rather than convergence on the structure of the external world. This study extends previous work by investigating direct alignment between vision and audio models without explicit cross-modal sensory training or linguistic intermediary, using 620 semantically paired samples from the AudioCaps 2.0 dataset. Layer-wise embeddings are extracted from DINOv2-large and DINOv2-base for vision, BEATs-iter3 and BEATs-iter3+ for audio, and Qwen3-1.7B for language. Alignment is measured using representational similarity analysis (RSA) and linear predictivity (LP) across all layer-embedding pairs and model combinations. Two controlled comparisons additionally test the effects of vision model capacity and audio pre-training supervision type, and two image construction approaches that include Stable Diffusion XL-generated and naturalistic video middle frames are compared to assess the effect of image type on alignment scores.

Results demonstrate significant vision–audio alignment emerging in mid-to-late layers, though alignment scores are consistently lower than both vision–language and audio–language alignment across all model combinations and metrics. Supervised audio models outperform self-supervised audio models, and generated images show stronger alignment earlier across layers than naturalistic middle frames, though both approaches reach comparable scores in later layers, with middle frames marginally outperforming generated images in the final layers for the supervised audio model. Vision model capacity shows only modest and inconsistent ef-

fects on alignment.

These findings support cross-modal convergence between independently trained vision and audio models while positioning language as the stronger representational bridge. This raises broader questions about whether convergence reflects language-like semantic abstraction, direct sensory world structure, or a mixture of both.

Code available [here](#).

1 Introduction

A central question for both cognitive neuroscience and machine learning is how distinct sensory systems develop internal representations of the world. In biological systems, vision, hearing, and other senses rely on unique neural pathways, yet they produce a unified perceptual experience. A similar question has emerged in deep learning about whether artificial neural networks trained on different modalities can independently arrive at similar representations, and if so, where and how does this convergence occur.

Recent research has explored representational convergence across artificial intelligence systems trained on different modalities, such as language, vision, and audio. Huh et al. (2024) proposed the “platonic representation hypothesis,” which suggests that neural networks trained on different tasks and modalities converge toward similar internal representations. Inspired by Plato’s notion of an underlying ideal reality, this hypothesis posits that models learn shared structural features of the world despite differences in training data and objectives. Using mutual k-nearest neighbors metrics, Huh et al. (2024) showed that representations in vision models align with those in language models, and this alignment increases with model scale and

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performance. These findings suggest that higher-performing models tend to converge toward a common representational structure.

Building on this idea of convergence, He et al. (2025) examined representational alignment specifically between vision and language models. Using layer-wise linear predictivity scores, they found that alignment between visual and linguistic representations peaks in middle to late layers of transformer architectures, rather than early layers (He et al., 2025). This suggests that early layers process modality-specific features, such as raw visual or linguistic patterns, while deeper layers develop shared semantic representations. Moreover, this alignment exhibits semantic sensitivity, remaining strong under changes to low-level visual features but weakening when high-level semantic content changes. He et al. (2025) also found that aggregating embeddings by averaging representations across multiple captions or images improves alignment, indicating a modality-independent semantic core underlying both vision and language representations.

Wang et al. (2025) further revealed that large language models can exhibit modality-specific representations through prompt engineering alone, without explicit training on visual or auditory data. Prior work understood representational alignment to emerge passively from training (Huh et al., 2024; He et al., 2025), but Wang et al. demonstrated that alignment can be actively steered at inference time through targeted prompting. By instructing models to "see" or "hear" inputs, Wang et al. generated representations that aligned with vision and audio model embeddings, measured with mutual k-nearest neighbors. A cautious interpretation of these results is that large language models encode linguistic and semantic regularities associated with perceptual experience, and that sensory prompts can steer their representations toward structures that better align with modality-specific encoders. These findings establish language models as useful benchmarks for understanding cross-modal alignment, and raise the question of how directly vision and audio models align with each other when language is removed as an intermediary.

While research has established alignment between language and other modalities, a critical gap remains in understanding direct alignment between vision and audio models. Language is already a

semantic medium used to describe both visual and auditory events, meaning that alignment between language and other modalities may partly reflect shared human labels, descriptions, or semantic abstractions rather than convergence on the structure of the external world (Wang et al., 2025; Huh et al., 2024). In contrast, vision and audition are independent sensory channels with fundamentally different input statistics and computational constraints. If independently trained vision and audio models show aligned representations for semantically paired events, this would provide stronger evidence that models are converging on shared structure in the world itself, rather than merely aligning through language-mediated semantics (Huh et al., 2024), providing a more direct test of the platonic representation hypothesis.

The present study addresses this gap directly by examining layer-wise cross-modal alignment among vision, audio, and language models using a controlled dataset of semantically paired audio clips and images from AudioCaps 2.0 (Wang et al., 2025). Representations are extracted using BEATs-iter3 and BEATs-iter3+ (Chen et al., 2023) for audio, DINOv2-large and DINOv2-base (Oquab et al., 2024) for vision, and Qwen3-1.7B (Yang et al., 2025) for language. Alignment is measured using representational similarity analysis (Kriegeskorte et al., 2008) and linear predictivity (Yamins et al., 2014) across all model combinations and layer pairs.

Two controlled comparisons isolate the effects of model capacity and pre-training supervision type on cross-modal alignment, and two image construction approaches are compared to assess the effect of image type on alignment scores. One image construction uses the Stable Diffusion XL model (SDXL) to generate synthetic images based on the audio captions in the dataset, and the other uses naturalistic middle frames extracted from the video sources tied to the same audio caption and sound. Additionally, a three-way comparison across all modality pairs examines the relative magnitude of vision-audio, vision-language, and audio-language alignment. Rather than testing whether language mediates sensory alignment, this comparison addresses a broader question of whether cross-modal convergence between vision and audio reflects language-like semantic abstraction, direct structure in the sensory world, or a mixture of both. Higher vision-language and audio-

language alignment relative to vision-audio would suggest that shared semantic descriptions partly drive representational similarity, while stronger vision-audio alignment in the absence of any linguistic signal would indicate that independently trained sensory models can recover shared world structure directly. This study addresses the following research questions:

1. Do independently trained vision and audio models show representational alignment on semantically paired inputs, and does this alignment follow the early-to-late progression observed in vision-language systems?
2. Is vision-audio alignment stronger or weaker than vision-language and audio-language alignment?
3. Does vision model capacity affect cross-modal alignment, such that larger models show stronger convergence with audio representations?
4. Do audio models trained with labeled data show stronger cross-modal alignment than those trained without labels?
5. Do images generated from expanded audio captions show greater alignment with audio than naturalistic images?

2 Methods

2.1 Dataset

This study uses the Audiocaps 2.0 dataset (Wang et al., 2025), which contains approximately 98,606 audio clips sourced from YouTube paired with natural language captions. Raw video and audio files were requested from the AudioCaps team, who provided 92,724 paired files. The filtered working dataset consists of 620 randomly selected samples with valid audio of 8 seconds or more, with a mean duration of 9.9 seconds. Clips longer than 10 seconds are trimmed to 10 seconds, and all clips are converted to 16kHz mono WAV before feature extraction.

2.2 Image Construction

Two parallel approaches to constructing semantically paired images are explored to compare align-

Caption: “a baby babbles while a girl and woman speak”

Expanded Caption: “A soft spotlight casts a gentle glow, illuminating a contented infant, who clutches a plush toy, emitting bubbles of curiosity. Amidst the ambient light, a girl and woman, with textured fabric patterns on their clothing, engage in warm conversation, the richness of their voices contrasting the baby”



Figure 1: Middle frame image



Figure 2: Generated image

ment scores across image types rather than individual embeddings.

SDXL-Generated Images. For each audio clip, Phi-3-mini-4k-instruct (Abdin et al., 2024) is used to expand the original AudioCaps caption into a richer visual scene description of approximately 80 tokens to emphasize the visual context of the caption before image generation. The LLM itself is prompted to maintain the audio description but add more visual context to the caption, while keeping the expanded caption under 60 words. One photorealistic image per sample is then generated using Stable Diffusion XL (Podell et al., 2023) with 40 denoising steps, a guidance scale of 8.5, and a fixed random seed (42) for reproducibility. The Stable Diffusion model itself contains a prompt that is prefixed with “Generate an image of:” and appended with photorealism descriptors including terms such as photorealistic, highly detailed, and sharp focus to guide the image generation.

Middle Frame Extraction. As a comparison approach, the middle frame is extracted from source video files received from the AudioCaps team, providing naturalistic real-world images directly corresponding to the audio content. Frames containing blank images, animations, video game content, and on-screen text or graphics are manually filtered out. The lower resolution of middle frames relative to SDXL images is not expected to substantially impact later-layer embeddings, as semantic content rather than resolution drives alignment in deeper transformer layers. An example of both extraction methods based on the caption is presented below.

2.3 Models

Two controlled comparisons are included to isolate specific factors contributing to cross-modal alignment, targeting model capacity in the vision models and pre-training supervision type in the audio models. All modalities are compared against a language model baseline using Qwen3-1.7B, allowing a full three-way comparison of vision, audio, and language representational alignment.

DINOv2-large and DINOv2-base (Oquab et al., 2024) are Vision Transformers pre-trained via self-supervised learning on a curated dataset of 142 million images, used for producing high-quality general-purpose visual features that transfer well across domains. DINOv2-base is included as a controlled comparison to isolate the effect of model capacity on cross-modal alignment, directly testing the platonic-representation hypothesis prediction that larger models show stronger representational convergence. Image representations are extracted from all hidden states using mean pooling over all patch tokens, where layer 0 corresponds to the patch embedding projection and subsequent layers correspond to successive transformer block outputs. Mean pooling is used instead of the CLS token to capture distributed spatial information from the entire image rather than a single summary token, and for consistency across all models in this study.

BEATs-iter3+ and BEATs-iter3 (Chen et al., 2023) are audio transformers pre-trained on AudioSet, a large-scale dataset of over 2 million 10-second audio clips spanning 527 sound event categories, using an iterative strategy in which an acoustic tokenizer and audio encoder are trained alternately. BEATs was chosen because it is

designed for general audio understanding rather than speech recognition, making it better suited for audio-visual alignment. BEATs-iter3 is included as a controlled comparison to assess the effect of pre-training supervision type, as it uses a self-supervised teacher trained without human-annotated labels, while BEATs-iter3+ uses a supervised fine-tuned teacher trained with AudioSet classification labels. Audio representations are extracted via PyTorch forward hooks on all 12 transformer blocks, with mean pooling applied over the time dimension. All embeddings are z-score normalized per layer before to ensure that our analysis reflects the structure of the embeddings, not arbitrary differences in scale.

Qwen3-1.7B (Yang et al., 2025) is a decoder-only large language model included to enable a complete three-way comparison of vision, audio, and language representational alignment. Wang et al. (2025) evaluate Qwen3 models across versions ranging from 0.6B to 32B on the AudioCaps dataset (Kim et al., 2019). Qwen3-1.7B is selected for direct comparability with their results while remaining computationally feasible. Original AudioCaps captions are used as input rather than Phi-3-expanded captions, to ensure the language model receives naturalistic text descriptions rather than synthetically enriched prompts. Layer-wise embeddings are extracted from all 29 transformer layers using mean pooling over the token dimension.

2.4 Computing Alignment

Cross-modal alignment is measured using two metrics, Representational Similarity Analysis and linear predictivity, applied to all modality pairs across all layer combinations

Representational similarity analysis (Kriegeskorte et al., 2008) is used to compare representational geometry across modalities without requiring a shared embedding space. For each combination of model layers, a Representational Dissimilarity Matrix (RDM) is constructed using correlation as the distance metric. RDMs are compared using Spearman rank correlation between their upper triangles, and this procedure is applied across all layer combinations to produce alignment heatmaps that reveal where in the network hierarchy cross-modal convergence emerges.

Linear predictivity provides a complementary metric to RSA to measure how well representations from one modality linearly predict those of

Table 1: Model architectures and embedding shapes

Model	Modality	Params	Layers	Hidden Dim	Tokenizer	Output Shape
DINOv2-large	Vision	307M	24	1024	Self-supervised	(25, 1024)
DINOv2-base	Vision	86M	12	768	Self-supervised	(13, 768)
BEATs-iter3+	Audio	90M	12	768	Supervised	(12, 768)
BEATs-iter3	Audio	90M	12	768	Self-supervised	(12, 768)
Qwen3-1.7B	Language	1.7B	29	2048	—	(29, 2048)

another. Ridge regression with cross-validated regularization is used to map embeddings across all layer pairs, estimated in both directions and averaged. Alignment is reported as the average pairwise correlation between predicted and true feature values across test sets in a 5-fold cross-validation scheme. While RSA measures similarity in representational geometry, linear predictivity tests whether that similarity is strong enough to support direct prediction across modalities.

Both metrics are applied to all three modality pairs: vision-audio, vision-language, and audio-language, using identical procedures across the same 620-sample dataset. For vision-audio comparisons, all models are evaluated on both generated images and middle frames, separately. For language comparisons, original AudioCaps captions are provided as input to Qwen3-1.7B to produce text representations that are compared against both vision and audio embeddings. This three-way comparison directly tests whether vision-audio alignment is stronger or weaker than alignment with language, situating the present findings within the broader landscape of cross-modal representational convergence.

3 Results

3.1 Vision-Audio Representational Alignment

Both RSA and linear predictivity analyses revealed consistent cross-modal alignment between independently trained vision and audio models on semantically paired inputs, addressing the primary question of whether such convergence exists without any shared training signal. Alignment was observed across all four vision-audio model combinations (DINOv2-large x BEATs-iter3, DINOv2-large x BEATs-iter3+, DINOv2-base x BEATs-iter3, DINOv2-base x BEATs-iter3+) and for both

image construction approaches, with peak linear prediction reaching an average pairwise correlation $r = 0.291$ (DINOv2-large x BEATs-iter3+, generated images) and $r = 0.295$ (DINOv2-large x BEATs-iter3+, middle frames). Layer-wise heatmaps from both analyses revealed that vision-audio alignment does not follow a simple monotonic early-to-late progression. Rather than peaking sharply in the deepest layers as has been reported for vision-language systems, alignment was distributed across mid-to-late layers of both architectures, with later BEATs layers (layers 8-11) showing the strongest alignment with later DINOv2 layers (layers 18-24 for large, layers 8-12 for base). Early vision layers showed consistently weaker alignment with all audio layers, consistent with the interpretation that low-level visual features have no acoustic counterpart. Together, these findings support a qualified version of the early-to-late progression hypothesis: cross-modal convergence does emerge in deeper layers, but the pattern is more diffuse across both architectures than what has been reported in vision-language comparisons.

3.2 Three-Way Comparison: Vision-Audio, Vision-Language, and Audio-Language Alignment

The linear predictivity scores show that vision-language alignment peaked at $r = 0.402$ followed by audio-language alignment ($r = 0.357$). In contrast, representational similarity analysis indicates that the structure of audio embeddings is more closely aligned with language compared to vision embeddings ($\rho = 0.290$ versus $\rho = 0.263$, $\Delta = 0.027$). The overall weakest pair was vision-audio alignment with $r = 0.295$ and $\rho = 0.243$.

Peak alignment for vision-audio and audio-language comparisons occurred in late layers of both architectures, consistent with the mid-to-late convergence pattern described in Section 3.1.

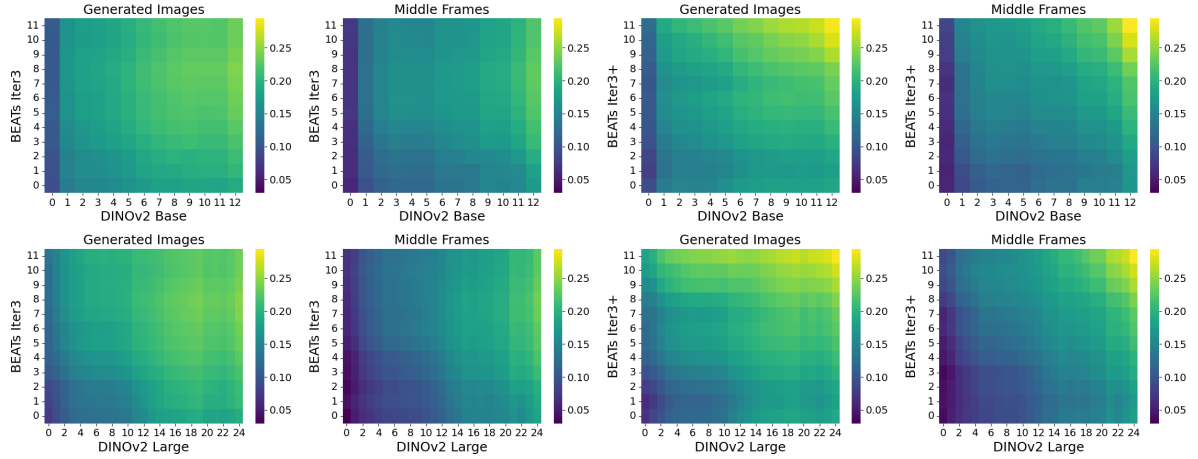


Figure 3: Vision Audio Linear Predictivity

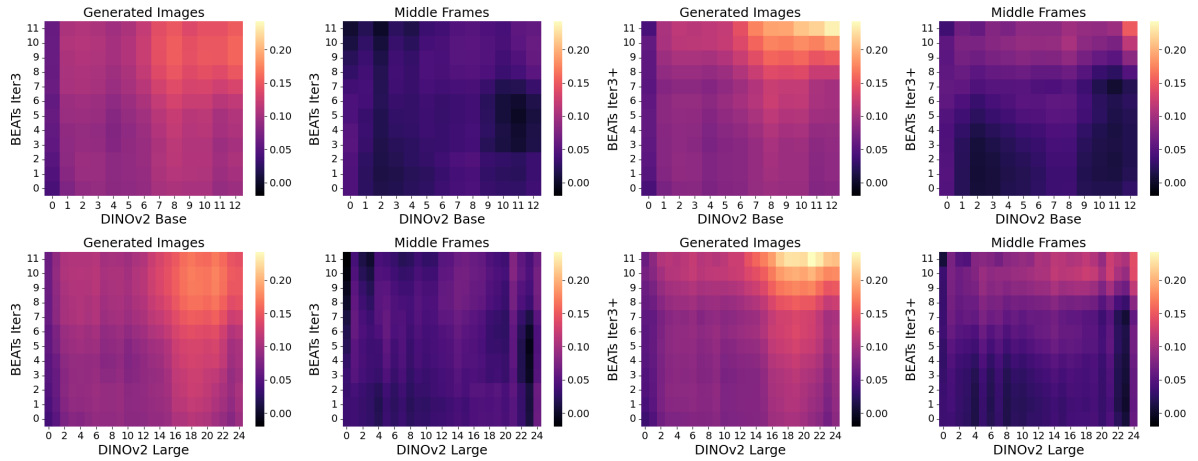


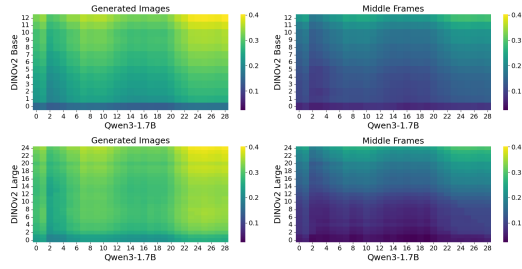
Figure 4: Vision Audio RSA

Vision-language RSA showed an exception, peaking at DINOv2-large layer 22 paired with Qwen3-1.7B layer 1 rather than a late language layer. This may reflect properties of Qwen3’s decoder-only architecture, in which early layers encode broad distributional structure that overlaps with patch-level visual features, though this pattern warrants further investigation.

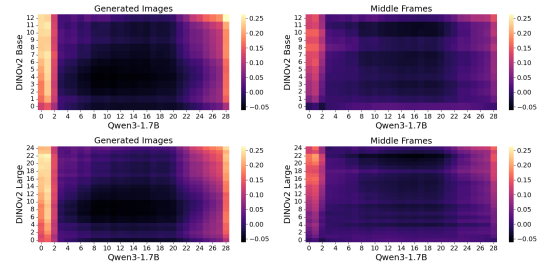
Despite the higher overall performance of language alignment results, vision-audio alignment was meaningful, with both metrics returning values systematically above chance across all model combinations, contesting a strictly language-centric account of cross-modal convergence while still positioning language as the stronger representational bridge.

3.3 Effect of Vision Model Capacity

The effect of vision model capacity was inconsistent across audio model pairings. DINOv2-large modestly outperformed DINOv2-base with BEATs-iter3 ($r = 0.236$ versus $r = 0.235$, $\Delta = +0.001$), but DINOv2-base and DINOv2-large performed equally well with BEATs-iter3+ for the final layers ($r = 0.291$, $\Delta = 0$). In both cases, the differences between model performance were small and did not hold consistently between the pairings of audio models. Peak alignment occurred in late layers for both model sizes, BEATs layers 10-11 and DINOv2 layers 21-24 for large and layer 12 for base, indicating that both model sizes converge on shared semantic structure in deeper representations. The absence of a consistent capacity advantage suggests that within the range tested, the size of vision models does not reliably amplify vision-audio convergence, and the platonic repre-

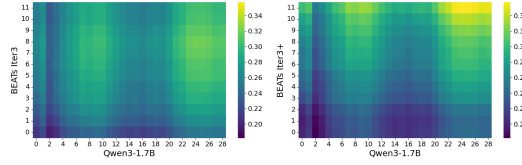


(a) Vision Language Linear Predictivity

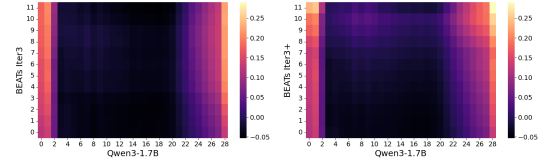


(b) Vision Language RSA

Figure 5: Vision Language Alignment



(a) Audio Language Linear Predictivity



(b) Audio Language RSA

Figure 6: Audio Language Alignment

sensation hypothesis prediction that larger models show stronger alignment is not clearly supported for this modality pair.

3.4 Effects of Audio Pre-Training Supervision

BEATs-iter3+, trained with supervised AudioSet classification labels, consistently outperformed BEATs-iter3 across both DINOv2 model sizes and both image types. With DINOv2-large, peak LP was $r = 0.291$ for iter3+ versus $r = 0.236$ for iter3 ($\Delta = +0.055$). With DINOv2-base, the advantage was comparable: $r = 0.291$ for iter3+ versus $r = 0.236$ for iter3 ($\Delta = +0.056$). The layer-wise profile was similar between the two models, with both showing strongest alignment in late layers, but iter3+ produced consistently higher scores throughout. The results are consistent with the idea that supervised AudioSet classification labels during pre-training produce audio representations that are more aligned with visual representations, though the analysis does not directly establish whether this reflects more semantically organized audio embeddings or other properties of the supervised training signal. Directly characterizing the internal structure of the audio representations would require additional analyses, such as testing alignment with category or event labels, which is a direction for future work.

3.5 Effects of Image Type on Vision-Audio Alignment

SDXL-generated images produced higher LP scores than naturalistic middle frames for the BEATs-iter3 model, with peak alignment reaching $r = 0.24$ for generated images with BEATs-iter3 and DINOv2-large, and $r = 0.23$ for middle frames under the same model configuration. As shown in Figure 3, generated images tend to reach higher alignment in earlier layers than middle frames for both BEATs-iter3 and BEATs-iter3+. For the BEATs-iter3+ comparison with DINOv2-large, alignment in the final layer is slightly higher for middle frames ($r=0.295$ versus $r=0.291$), though this difference is marginal.

These results should be interpreted carefully, as the two image types differ not only in visual quality but in what they represent. Generated images and conditioned directly on expanded AudioCaps captions, producing a standardized visual realization of the event description. Middle frames, by contrast, are temporarily sampled from the source video, and a single frame may not fully summarize the audio event. The sound source may be off-screen, occur before or after the middle frame, or be only partially visible. Therefore, higher alignment for generated images does not necessarily indicate that synthetic images are superior visual stimuli for cross-modal alignment research. Rather, it may reflect that caption-conditioned gen-

erated images are more semantically standardized and more directly aligned with the caption-level event description, whereas naturalistic middle frames introduce temporal and spatial variability that reduces alignment even when the semantic pairing is valid.

The example images in Figures 1 and 2 in Section 2.2 illustrate this distinction. The generated image directly depicts the woman and baby described in the caption, while the middle frame shows only individual limbs of children holding a book. This is a real sample from the scene, but it does not capture the semantic content of the caption. This contrast highlights a fundamental difference between caption-aligned generated visualizations and temporally sampled natural video frames, rather than a straightforward quality advantage for synthetic images. Future work should consider this distinction when constructing visual stimuli for alignment studies, and could explore alternative frame selection strategies or multiple frames per clip to better capture the event described by the audio caption.

4 Discussion

This study examined layer-wise cross-modal representational alignment among independently trained vision, audio, and language models on semantically paired inputs, contributing new evidence to the literature on representational convergence in deep learning.

4.1 Video-audio Alignment and early-to-late progression.

Vision and audio models converge on shared representational structure without any joint training signal, with alignment emerging consistently in mid-to-late layers across all four model combinations and both image types. This broadly reproduces the early-to-late pattern He et al. (2025) described for vision-language systems, though the vision-audio pattern is more spatially diffuse, suggesting that vision and audio arrive at shared structure through a more distributed set of perceptual regularities than the sharper late-layer convergence observed in vision-language systems.

4.2 Language as a stronger representational bridge.

The greater alignment metrics for vision-language and audio-language compared to vision-audio alignment positions language as a stronger representational bridge than direct perceptual co-occurrence between vision and audio. The particular strength of audio-language alignment is consistent with Wang et al. (2025), who demonstrated that Qwen3 models encode auditory information through text-based representations, suggesting that co-occurrence of text with sound descriptions during training may produce representations that are especially align-able with audio. Despite the ordering, vision-audio alignment returned values systematically above chance across all combinations, contesting a strictly language-centric account of cross-modal convergence.

4.3 Model capacity.

The inconsistent effect of vision model capacity, with DINOv2-large slightly outperforming base with BEATs-iter3 but equaling performance with BEATs-iter3+, suggests that within the range tested, model size does not reliably drive vision-audio convergence. This provides limited support for the platonic-representation hypothesis that larger models show stronger representational alignment, at least for this modality pair.

4.4 Supervision type.

The consistent advantage of BEATs-iter3+ over BEATs-iter3 across both vision model sizes indicates that AudioSet classification labels during audio pre-training improve cross-modal alignment with vision. This contrasts with findings in the vision literature suggesting that self-supervised models can match supervised ones on representational geometry tasks (Oquab et al., 2024), and suggests that explicit semantic supervision may be more consequential for audio than for vision when measuring cross-modal alignment.

4.5 Image type.

The advantage of generated images over naturalistic middle frames across all model combinations inverts the assumption that naturalistic inputs are

preferable for alignment studies. Generated images conditioned on AudioCaps captions may produce more semantically regular visual representations that are more linearly predictable from audio embeddings, whereas naturalistic frames introduce visual variability in framing, lighting, and scene context that reduces linear alignment even when the semantic pairing is valid. This has practical implications for how future alignment research constructs visual stimuli and further suggests that the choice of image source can substantially affect alignment estimates.

5 Limitations & Future Directions

Sample size. The dataset consists of 620 paired audio-image samples, which is relatively small compared to other alignment analyses. Given the 80/20 train/test split used for linear predictivity, each fold yields approximately 80 training samples, which may reduce the reliability of linear predictivity estimates and limit the generalization of findings. While prior work suggests that sample size is less critical for RSA than for linear predictivity, future work should evaluate whether alignment patterns are consistent with larger sample sizes.

Model selection. This study examines alignment using two vision models and two audio models, which represents only a subset of available models. While the models were selected for their strong benchmark performance and general-purpose feature quality, other architectures, training objectives, or model families may produce different alignment patterns. Specifically, results may not generalize to convolutional architectures or models trained on different types of data. Similarly, only one language model, Qwen3-1.7B, is included in the three-way comparison, and alignment patterns with language may vary across model families, size, and training objectives. Future work should examine whether the observed alignment patterns replicate across a broader range of models.

Image generation. Both Stable Diffusion XL model and Microsoft Phi-3-mini-4k instruct model have architectural and performance limitations that can affect image generation quality. Phi-3-mini-4k instruct, used to expand the captions, was prompted to keep expanded captions under 60 tokens and to not include instructions in the out-

put, yet the model failed to follow formatting instructions by returning the instructions alongside the new captions and generating captions that exceeded the token constraint. Additional filtering was required to isolate the expanded captions before image generation. However, some captions remained longer than 77 tokens, cause the SDXL text encoder, CLIP-ViT-L/14, to clip those captions. This may have affected the image quality, as these expanded captions served as the primary guide for what visual characteristics were to be included in the generated image.

Additionally, SDXL tends to struggle architecturally with attribute leakage, particularly in its cross-attention mechanism with processing multiple CLIP tokens. This can cause the model to merge distinct subjects into one rather than generating them separately. As shown in **Figure 7** below, instead of generating a terrier dog and a deer as two separate animals, the model produced a hybrid image of a terrier-deer mix.

Expanded Caption: “A bristled, terrier-sized dog stands alert, its coat a muddy brown against the lush green brush. Sunlight filters through the canopy, casting dappled shadows on the scene. A shadowy figure emerges—a sturdy deer, its antlers twisted from a recent encounter with a branch.”

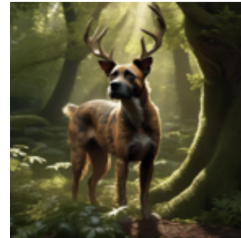


Figure 7: Hallucinated image of dog-deer

This type of error can distort the semantic correspondence between the generated image and the audio caption, potentially reducing alignment scores for affected samples. SDXL is also computationally expensive, requiring smaller batch sizes and limiting the total number of samples that can be generated efficiently. Future directions to address these limitations could include fine-tuning the LLM on a curated dataset of audio captions tailored to the task at hand, or trying structured prompting strategies such as Chain of Thought reasoning to explain its process in expanding captions to limit further hallucinations.

Middle frame filtering. The extracted middle

frames were manually filtered to remove images with on-screen text/graphics, blank screens, and animations or video-game screenshots. As a result, alignment estimates may become biased toward naturalistic image representations. Future work could remove or reduce filtering restrictions before computing alignment to improve ecological validity since real-world image streams frequently contain overlays, subtitles, and other elements.

6 Conclusion

This study demonstrates that independently trained vision and audio models develop meaningful representational alignment on semantically paired inputs, with such convergence emerging primarily in mid-to-late layers. However, direct vision–audio alignment remains the weakest of the three modality pairs, consistently falling below both audio–language and vision–language alignment. These results support the Platonic Representation Hypothesis for perceptual modalities while demonstrating the role of language as a cross-modal bridge. Additional findings reveal that semantically conditioned synthetic images enhance alignment in the earlier layers, but both middle frames and generated images reach similar alignment results in the end. While vision model capacity yields only modest benefits for alignment, an improved, pre-trained audio model (Beats-iter3+) does have a significant effect on alignment results, as the middle frames and generated images generate much higher alignment results in the late layers. Together, these results challenge the assumption that naturalistic stimuli, such as middle frames, are optimal for alignment, and instead suggest new practical considerations for multi-modal convergence.

By clarifying how vision, audio, and language representations relate, this work advances understanding of emergent structure in deep neural networks and opens up opportunities for more effective multi-modal model design.

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